Two-Failure Prefix Summaries in Oracle Matroids

Yilin Liu

26 October 2026

Abstract

Fix an order on a rank-r matroid and assign at most one failure label to each item. We study the unique minimum subset preserving every ordered-prefix rank after at most two label failures. Its worst-case size is exactly $3r$ through rank three and exactly 13 at rank four. General upper bounds follow from one-failure replacement profiles and a circuit missing-pair argument. If the one-failure union adds only one new item, appearing under t single failures, the sharp size is $r + 2t$. We give a profile-sensitive construction and a 13-item nonrepresentable obstruction. The proofs allow loops, unlabelled items, rank drops and arbitrary fixed signed costs.

A direct construction. Let N be the number of items. Compute the canonical basis B_0. For each label on B_0, compute the basis after that label fails; for each label on each resulting basis, compute the basis after both labels fail. Retain their union. For a target failure set, a basis avoiding all remaining failed labels is unchanged by those deletions. Otherwise at least one failed label occurs on that basis and is a branch of the construction. Following at most two such branches therefore reaches the target canonical basis. Every queried set is itself allowed, so the retained union is exactly the union over all allowed failures. There are at most $1 + r + r^2$ scans and at most $N(1 + r + r^2)$ independence queries after fixing the order. The proofs below establish its size, uniqueness and improved structured case.

Part I

Profile bounds and sharp structured cases

I.1 Scope and results

Let M be an arbitrary matroid of rank r on a finite explicitly listed ground E, with one fixed total item order. Each item has at most one label; unlabelled items are allowed. A failed label deletes its entire class. Let H_j be the union of the canonical greedy bases after at most j label failures, including the unfailed query.

The order may be nondecreasing signed rational cost followed by a permanent item ID. Every statement concerns that one fixed order. The general prefix theorem shows that H_j is the unique smallest original-item set preserving every surviving prefix rank for those queries. It therefore preserves the canonical minimum-cost surviving bases, including rank drops.

This note proves the following bounds for **two failures and at most one label per item**:

$$|H_2| \le \min\left\{5r, \binom{r+1}{2} + 3\right\} \qquad (r \ge 2). \tag{TF1}$$

The bounds at ranks zero and one are zero and three, respectively. An instance-dependent bound from the one-failure replacement profile is stronger and is stated below. It gives the sharp $3r$ bound whenever no replacement item appears in more than two single-failure bases. The same conclusion holds whenever all single failures add only one shared replacement item, even if its multiplicity is large (Section 5). In particular,

$$\boxed{|H_2| \le 3r \quad \text{for every matroid of rank } r \le 3.} \tag{TF2}$$

At rank four the universal bound is 13, and the explicit obstruction attains it. Thus the largest possible size of the minimum prefix preserver is **exactly 13 at rank four**. The same $3r$ bound holds at every rank when at most three labels occur. Together these facts prove that the rank-four, four-label, 13-item obstruction has the smallest possible rank, label count and original ground size for violating the represented $3r$ bound at these fixed failure parameters. This is a mathematical minimality statement within this model, not a publication-priority claim or a statement for other failure budgets.

The ordered-basis tree constructs H_2 with at most $N(1+r+r^2)$ independence queries, where $N = |E|$, by rescanning the fixed ordered input at each tree node. This is $O(N(r+1)^2)$ queries after sorting. The retained-prefix witness alternative scans that input once and uses at most $N(K+1)(1+r+r^2)$ queries, where K is any valid bound on $|H_2|$, clipped at N. At any fixed rank both bounds are linear in N. No representation or field arithmetic is needed.

I.2 The single-failure replacement profile

Let B_0 be the ordinary canonical basis. For each occurring label λ, let D_λ be its class and B_λ the canonical basis of $E \setminus D_\lambda$. Define

$$R_\lambda = B_\lambda \setminus B_0, \qquad G = H_1 \setminus B_0, \qquad t_e = |\{\lambda : e \in R_\lambda\}| \quad (e \in G),$$

$$T = \sum_{e \in G} t_e = \sum_{\lambda} |R_\lambda|. \tag{TF3}$$

Thus t_e counts the single-failure bases in which the same replacement item appears; T counts these appearances with multiplicity. Every t_e is positive. A label absent from B_0 leaves that basis unchanged, so only at most r labels need to be queried. The entire profile can be computed with at most $(1+r)N$ independence queries.

Let b_0 be the number of unlabelled items in B_0, and let

$$\delta_\lambda = r - r_M(E \setminus D_\lambda) \tag{TF4}$$

be the rank lost under the single failure. Greedy-accepted items persist under deletions that keep them. Consequently B_λ contains $B_0 \setminus D_\lambda$, and

$$|R_\lambda| = |B_0 \cap D_\lambda| - \delta_\lambda.$$

The label classes are disjoint, so

$$T = r - b_0 - \sum_{\lambda} \delta_\lambda \le r. \tag{TF5}$$

This identity allows loops, unlabelled items and rank drops. It does not assume that every surviving ground has full rank.

I.3 Profile bound with rank losses included

Theorem 1 (replacement-profile bound). With the notation above,

$$\boxed{|H_2| \le r + 2T + \sum_{e \in G} \binom{t_e - 1}{2}.} \tag{TF6}$$

Here $\binom{0}{2} = \binom{1}{2} = 0$. There is also the bound

$$|H_2| \le r + 4T. \tag{TF7}$$

Proof. For every failed label set F, including sets larger than two, rank submodularity gives

$$r_M(E_F) \le r - \sum_{\lambda \in F} \delta_\lambda. \tag{TF8}$$

Indeed, when $\lambda \notin F$, the sets $E \setminus D_\lambda$ and E_F have union E and intersection $E_{F \cup \{\lambda\}}$. Submodularity says that deleting λ after F loses at least δ_λ rank. Add these inequalities one failed label at a time. This step uses disjoint label classes; overlapping lists of two or more labels do not satisfy the same argument.

Independently fail each label with probability $p \in (0,1)$ and put $q = 1 - p$. By TF8, the expected surviving rank is at most $r - p\sum_\lambda \delta_\lambda$. The items of B_0 contribute exactly $b_0 + (r - b_0)q$ to expected greedy rank.

An item $e \in G$ is accepted whenever any one of its t_e witness labels fails and its own label, if present, survives. These are disjoint label sets. Its acceptance probability is therefore at least $q(1 - q^{t_e})$. For an unlabelled item, omitting the factor q would give an even stronger probability, so the common lower bound is valid.

Put $n_2 = |H_2 \setminus H_1|$. Each of these remaining items has a two-failure prefix witness and contributes at least qp^2 to expected rank. Deleting additional earlier items cannot invalidate acceptance of a surviving protected item. Combining the disjoint item contributions with TF5 gives

$$n_2 q p^2 \le pT - q \sum_{e \in G} (1 - q^{t_e}). \tag{TF9}$$

For every fixed positive integer t,

$$q(1 - q^t) = tp - \left(t + \binom{t}{2} \right) p^2 + O(p^3) \quad \text{as } p \downarrow 0.$$

Since $\sum_e t_e = T$, divide TF9 by qp^2 and let $p \downarrow 0$ to obtain

$$n_2 \le T + \sum_{e \in G} \binom{t_e}{2}.$$

Add $|H_1| = r + |G|$ and rearrange each summand to get TF6.

For TF7, ignore the extra one-failure information and give every item of $H_2 \setminus B_0$ its general lower probability qp^2. Subtracting the B_0 contribution from the upper expected rank leaves exactly pT, by TF5. Thus $|H_2| - r \le T/(pq)$; take $p = q = 1/2$. No random choices are used by the summary algorithm. The product measure only proves its size. □

If every $t_e \le 2$, the sum in TF6 vanishes, giving $|H_2| \le r+2T \le 3r$. This applies to arbitrary matroids satisfying that explicit profile condition, regardless of representability.

For a parameter-only bound, note that $g(t) = \binom{t-1}{2}$ satisfies $g(a+b)-g(a)-g(b) = ab-1 \ge 0$ for positive integers a, b. If $T > 0$, then $\sum_e g(t_e) \le g(T)$. Since $T \le r$, TF6 gives

$$|H_2| \le r + 2r + \binom{r-1}{2} = \binom{r+2}{2} \quad (r \ge 1).$$

If $T = 0$, TF6 gives $|H_2| = r$ directly; rank zero is also immediate. This proves the preliminary bound $|H_2| \le \min\{5r, \binom{r+2}{2}\}$, including $|H_2| \le 3$ at rank one. The next section improves the quadratic term.

I.4 A circuit argument improves the universal bound

I.4.1 Which missing pairs can a new item replace?

Lemma 2 (circuit extension). Let C be a circuit of size $m \ge 3$, and let $y \notin C$ be a nonloop in its span. Form a graph on C whose edges are the pairs $\{a, b\}$ for which

$$y \notin \mathrm{cl}_M(C \setminus \{a, b\}).$$

This graph is complete multipartite with at least two nonempty parts. Every induced subgraph on s vertices therefore has either no edges or at least $s-1$ edges.

Proof. Define $a \sim b$ if $a = b$ or if $y \in \mathrm{cl}_M(C \setminus \{a, b\})$. Symmetry and reflexivity are immediate. For distinct a, b, c, the flats

$$L_{ab} = \mathrm{cl}_M(C \setminus \{a, b\}), \qquad L_{bc} = \mathrm{cl}_M(C \setminus \{b, c\})$$

each have rank $m-2$. Their union has rank $m-1$: it contains the basis $C \setminus \{b\}$ and lies in $\mathrm{cl}_M(C)$. Rank submodularity bounds their intersection rank by $m-3$. Their intersection already contains the rank-$m-3$ flat $\mathrm{cl}_M(C \setminus \{a, b, c\})$. Since a proper inclusion between flats increases rank, the two flats are equal:

$$L_{ab} \cap L_{bc} = \mathrm{cl}_M(C \setminus \{a, b, c\}).$$

Thus $a \sim b$ and $b \sim c$ imply $a \sim c$, proving transitivity.

There cannot be just one equivalence class. Fix $c \in C$ and put $B = C \setminus \{c\}$. For subsets A, A' of the independent set B, rank submodularity similarly gives

$$\mathrm{cl}_M(A) \cap \mathrm{cl}_M(A') = \mathrm{cl}_M(A \cap A').$$

If all pairs were equivalent, y would belong to every $\mathrm{cl}_M(B \setminus \{a\})$ for $a \in B$. Repeatedly applying this equality puts y in $\mathrm{cl}_M(\varnothing)$, contradicting that it is a nonloop. Edges are exactly the pairs in different equivalence classes, so the graph has the asserted form. If an induced subgraph is nonempty, choose one of its parts of size a, where $1 \le a < s$. Its edges to the other vertices alone number $a(s-a) \ge s-1$. □

I.4.2 All but one replacement profile

We prove $|H_2| \le \binom{r+1}{2} + 3$ for $r \ge 2$. If $1 \le T \le r-1$, the preliminary convexity calculation gives

$$|H_2| \le r + 2(r-1) + \binom{r-2}{2} = \binom{r+1}{2} + 1,$$

and $T = 0$ gives the stronger bound r directly. If $T = r$ and $|G| \ge 2$, the sum of $g(t_e) = \binom{t_e - 1}{2}$ is maximized by multiplicities $r-1, 1$. To see this, the increments $g(t+1) - g(t) = t - 1$ are nondecreasing: moving a unit from a smaller part of size at least two to a largest part cannot reduce the sum. Keep at least two positive parts. TF6 then gives

$$|H_2| \le 3r + \binom{r-2}{2} = \binom{r+1}{2} + 3.$$

Only the following profile remains:

$$T = r, \qquad G = \{x\}, \qquad t_x = r. \tag{TF10}$$

I.4.3 The exceptional profile starts with a circuit

For any prefix P of the fixed order, write $H_j(P)$ for the canonical union computed in the restriction to P. Greedy decisions never change after a prefix has been scanned, so

$$H_j(P) = H_j(E) \cap P \quad (j = 1, 2). \tag{TF11}$$

If $H_1(P)$ is just its unfailed basis, its profile has $T(P) = 0$. TF6 then says that $H_2(P)$ is that same basis. In particular, no two-failure-only item precedes x in the exceptional case.

Each label whose failure makes x enter greedy must hit an earlier item of B_0; otherwise the earlier unfailed basis survives and continues to span x. Since $t_x = r$, all r items of B_0 precede x and have distinct labels. TF5 also gives $b_0 = 0$ and $\delta_\lambda = 0$ for every label.

We may restrict to H_2: it contains every required canonical basis, and deleting only rejected items preserves all required prefix ranks and greedy decisions. Indeed, induction through the order leaves the accepted set unchanged whenever a rejected item is removed. The first $r+1$ remaining items are therefore $Q = B_0 \cup \{x\}$. For each $a \in B_0$, failing its label gives the basis $(B_0 \setminus \{a\}) \cup \{x\}$. Together with B_0, these are all r-subsets of Q. Thus Q is a circuit of size $r+1$ and rank r.

The item x is either unlabelled or has a new label: it survives each of the r witness labels. If it is unlabelled, only the $\binom{r}{2}$ failed pairs among the labels of B_0 can remove two items of Q. Every other query leaves r independent items of Q and accepts no later item. Each relevant pair leaves rank $r-1$ inside Q and accepts at most one later item. Hence

$$|H_2| \le r + 1 + \binom{r}{2} = \binom{r+1}{2} + 1.$$

Assume that all $r+1$ items of Q have distinct labels. Only the $\binom{r+1}{2}$ failed pairs of these labels can accept a later item. Each query accepts at most one: the first surviving item outside the closure of its surviving $r-1$ items of Q. Call it the query's winner. If there is no later winner, the required bound already holds.

Otherwise let y be the first item of H_2 after Q. It is a nonloop and lies in the span of Q, since that circuit already has the full rank r. Lemma 2 says that the failed pairs for which it would increase rank form a complete multipartite graph on Q.

If y has the label of some $c \in Q$, it survives exactly the failed pairs that avoid c. Its winning pairs are therefore the edges in the induced graph on $Q \setminus \{c\}$. This graph has an edge because y

wins a query, so it has at least $r-1$ edges. If y has an outside label or is unlabelled, it survives all these pairs and wins at least r queries, which is stronger. There are no earlier later winners to compete with y in either case.

Thus one item accounts for at least $r-1$ of the relevant queries. All other queries together supply at most $\binom{r+1}{2} - (r-1)$ distinct later winners. Consequently

$$|H_2| \le (r+1) + 1 + \binom{r+1}{2} - (r-1) = \binom{r+1}{2} + 3.$$

Combining this with TF7 proves TF1. In particular the universal bounds at ranks two, three and four are 6, 9 and 13. The rank-one bound proved above completes TF2. Rank-r partition matroids with three ordered copies per parallel class, labelled by the same three distinct labels in every class, attain $3r$ under two failures. The explicit rank-four obstruction attains 13. These upper bounds use only matroid ranks.

I.5 One shared single-failure replacement has a sharp linear bound

Theorem 3 (one replacement item). Suppose $H_1 \setminus B_0 = \{x\}$. Let W be the set of labels whose single-failure canonical basis contains x, and put $t = |W| = T \ge 1$. Then

$$\boxed{|H_2| \le r + 2t \le 3r.} \tag{TF12}$$

The bound $r + 2t$ is attained by explicitly represented matroids for every $1 \le t \le r$. If x is unlabelled, there is the stronger bound

$$|H_2| \le r + 1 + \max\{0, 2t - 3\}.$$

No assumption is made about the number of items of B_0 sharing a label or about rank losses after a failure. If $H_1 = B_0$ instead, TF6 gives $H_2 = B_0$ directly.

I.5.1 Only a small set of failed pairs can add items

Write $b_\lambda = |B_0 \cap D_\lambda|$. The single-failure basis is $B_0 \setminus D_\lambda$, together with x exactly when $\lambda \in W$. Therefore

$$\delta_\lambda = b_\lambda - \mathbf{1}[\lambda \in W].$$

For every $|F| \le 2$, TF8 gives

$$r_M(E_F) \le |(B_0)_F| + |F \cap W|.$$

Every surviving original-basis item is still accepted. If x survives and $F \cap W \ne \varnothing$, deletion stability also keeps x accepted. Thus an item of $H_2 \setminus H_1$ can appear only after one of the following pairs: two labels of W, or a label of W together with the own label X of x, when x is labelled. The latter label is outside W. Each such pair accepts at most one item outside H_1.

We may move all original-basis items before the other items without changing any required canonical basis. For a fixed failure, its old accepted set already contains the promoted surviving basis items. Induction through the remaining order keeps that same independent accepted set: an old rejected item is still spanned by its old earlier accepted items, possibly with more

promoted items. The prefix argument in Section 4.3 shows that no item of $H_2 \setminus H_1$ precedes x. After promotion, the critical items are therefore B_0, then x, then the remaining items in their original relative order.

I.5.2 A circuit quotient preserves each relevant replacement test

Let C be the fundamental circuit of x with respect to B_0: the unique circuit in $B_0 \cup \{x\}$. For every $\lambda \in W$, choose $u_\lambda \in C \cap D_\lambda$. Such an item exists because otherwise $B_0 \setminus D_\lambda$ would span x. Set

$$A = B_0 \setminus \{u_\lambda : \lambda \in W\}, \qquad Q = \{u_\lambda : \lambda \in W\} \cup \{x\}.$$

Contract the independent set A for this proof. The contraction has rank t, and Q is a $(t+1)$-item circuit: deleting x from Q lifts to B_0, while deleting u_λ lifts to the basis $B_0 - u_\lambda + x$.

Some contracted items may have failed labels. The following explicit intersection argument is needed; treating them as surviving would be incorrect. For a relevant failed pair F, let $S_F = (H_1)_F$. For every surviving $e \notin H_1$,

$$e \notin \mathrm{cl}_M(S_F) \quad \Longleftrightarrow \quad e \notin \mathrm{cl}_{M/A}(Q_F). \tag{TF13}$$

First take $F = \{\lambda, \mu\} \subseteq W$, and write

$$Z = (B_0 \setminus \{u_\lambda, u_\mu\}) \cup \{x\}, \qquad L_\lambda = \mathrm{cl}_M((B_0 \setminus D_\lambda) \cup \{x\}),$$

with L_μ defined similarly. The item e belongs to both single-failure basis flats L_λ, L_μ. The ranks of $\mathrm{cl}_M(Z)$ and L_λ are $r-1$ and $r - b_\lambda + 1$. Their union has rank r because it contains $B_0 - u_\lambda + x$. Their intersection therefore has rank at most $r - b_\lambda$ and contains the independent set $(B_0 \setminus (D_\lambda \cup \{u_\mu\})) \cup \{x\}$ of that rank. Hence it is exactly that set's closure. Intersecting with L_μ similarly has rank at most $r - b_\lambda - b_\mu + 1$: the union in the submodularity inequality contains $B_0 - u_\mu + x$, of rank r. The intersection contains S_F of that exact rank, so

$$\mathrm{cl}_M(Z) \cap L_\lambda \cap L_\mu = \mathrm{cl}_M(S_F).$$

Since $Z = A \cup Q_F$, this proves TF13 for such pairs.

For $F = \{\lambda, X\}$, use $Z = B_0 \setminus \{u_\lambda\}$ and $L_X = \mathrm{cl}_M(B_0 \setminus D_X)$. The same calculation gives $\mathrm{cl}_M(Z) \cap L_\lambda = \mathrm{cl}_M(B_0 \setminus D_\lambda)$. Intersecting with L_X gives $\mathrm{cl}_M(B_0 \setminus (D_\lambda \cup D_X))$, because these are closures of subsets of the independent set B_0. This is $\mathrm{cl}_M(S_F)$ and proves TF13 in the remaining case. These calculations explicitly allow $b_\lambda > 1$ and $D_X \cap B_0 \neq \varnothing$.

I.5.3 A decreasing potential bounds all later winners

When x has label X, relevant failed pairs are the edges of a complete graph on $W \cup \{X\}$, with $n = t+1$ vertices. If x is unlabelled, use the complete graph on W, with $n = t$ vertices. Initially every such query is unresolved. Each later critical item wins at least one unresolved query and resolves every surviving query for which it increases rank.

By TF13 and Lemma 2 in the circuit quotient, its rank-increasing pairs form a complete multipartite graph, restricted to the relevant labels. The two-item circuit case has the same property directly. If its own label is one of those vertices, pairs incident with that vertex delete the item and cannot be won. Thus a winner removes all edges between different parts of a partition of the current graph, except that edges incident with one protected vertex may be spared.

The unresolved graph always decomposes into clique blocks joined only through separating vertices. Initially there is just one clique. An operation splits a block into disjoint cliques if its protected vertex is absent, or into cliques sharing just that vertex if it is present. These operations preserve the forest of blocks. Ignore isolated vertices, and use the potential

$$\Phi(G) = \sum_{\text{nontrivial blocks } D} (2|D| - 3).$$

A protected split into $k \geq 2$ nontrivial cliques sharing the protected vertex lowers the potential by $k - 1$. For an unprotected split with a singleton parts and b parts of size at least two, the decrease is $2a + 3b - 3$. It is at least one whenever an edge is removed. An operation may split several blocks, so every actual winner decreases Φ by at least one.

The initial potential is $2n - 3$ for $n \geq 2$ and zero for $n \leq 1$. Hence there are at most $\max\{0, 2n-3\}$ later critical items. Adding the $r+1$ items of H_1 proves TF12 and the unlabelled refinement. This graph is a proof device; the summary algorithm need not construct it or perform any contraction.

I.5.4 Sharp represented examples and faster construction

For $t \geq 2$, take the ordered vectors $e_1, \ldots, e_t$ of a standard basis over $\mathbb{F}_2$, labelled $1, \ldots, t$, followed by $x = e_1 + \cdots + e_t$ with a new label X. Append:

1. one copy of e_i labelled X for each $1 \leq i < t$;
2. one copy of e_i for each $1 \leq i \leq t$, labelled $i + 1$ cyclically modulo t.

The single-failure union adds only x, with multiplicity t. The first appended copy of e_i is forced by failures $\{i, t\}$: earlier appended copies are parallel to surviving initial basis items. The second appended copy of e_i is forced by $\{X, i\}$, which removes every copy in the first appended group. Thus all $3t$ items are necessary. Adding $r - t$ unlabelled coloops gives exactly $r + 2t$ forced items. For $t = 1$, use three parallel items with distinct labels and add $r - 1$ unlabelled coloops. Powers-of-two costs in the displayed order also force every item for fixed optimum-value preservation.

The profile gives an algorithmic improvement. Compute B_0 and its single-failure bases for the $\ell_0 \leq r$ labels touching B_0. Other labels leave B_0 unchanged. If the extra-item union is empty, return B_0. If it consists of x, compute just the relevant pair-failure bases described in Section 5.1 and add them to H_1. All other queries have no item outside H_1, so this returns exactly H_2. The independence-query bound is

$$N\left(1 + \ell_0 + \binom{t+1}{2}\right), \tag{TF14}$$

or $N(1 + \ell_0 + \binom{t}{2})$ when x is unlabelled. Thus the observable singleton profile permits $O(N(r + t^2))$ oracle work after sorting, while retaining the original oracle and all canonical prefix ranks. No field representation is required by this algorithm.

I.6 Three occurring labels also suffice at every rank

Let E_0 be the unlabelled core and $D_1, \ldots, D_L$ the nonempty classes, where $L \leq 3$.

If $L \leq 2$, every critical labelled item belongs to the canonical basis of E_0 together with its own label class: delete all other classes after a witness and use greedy stability. Every critical

unlabelled item belongs to the basis of E_0, obtained by deleting all L classes. These are at most $L + 1 \leq 3$ bases, so their union contains H_2 and has size at most $3r$.

If $L = 3$, use just the three canonical bases of $E_0 \cup D_i$. A critical labelled item belongs to the basis for its own label. A critical unlabelled item has a witness leaving at least one class D_i; extend that witness by deleting the other two classes. The item survives, is still accepted, and lies in the i-th basis. Again $|H_2| \leq 3r$.

I.7 Minimality and the exact rank-four maximum

The explicit obstruction has rank four, four occurring labels and 13 original items. Section 4 excludes a violation at any smaller rank, and Section 6 excludes a violation with fewer labels. Any violation must force at least $3r+1 \geq 13$ items, so the example also has the smallest possible original ground size.

Theorem 3 also shows that any violation needs at least two distinct replacement items in the one-failure union. The example has exactly two, so it attains this minimum as well.

The example's powers-of-two costs make the relevant optimum bases unique. Its 13 items are therefore necessary even if a restriction need only preserve the fixed optimum-cost values. The upper results in this note preserve all prefixes and hence imply the weaker optimum-value guarantees. The minimality statement covers both requirements at $h = 2, d = 1$; it does not address arbitrary cost queries or other failure budgets and label degrees.

For that particular instance, its unfailed basis is $\{1, 2, 3, 4\}$, and its only one-failure replacement items are item 5 with multiplicity three and item 6 with multiplicity one. Thus $T = 4$ and TF6 gives $4 + 8 + 1 = 13$, an exact upper certificate for the example. Section 4 proves that every rank-four instance has a prefix preserver of at most 13 items, so the rank-four maximum is exactly 13. The optimal general-rank constant remains open here: the direct sum of this example forces $13r/4$ items when four divides r, while the universal linear upper bound is $5r$.

I.8 Verification

The prefix checker includes a separate degree-one profile suite over all 92 labelled matroids on at most four items, with unlabelled items, loops and rank drops. It checks TF5, TF8 for every failure subset, TF6, TF7, and TF1 against independently enumerated canonical bases. It also checks the profile of the nonrepresentable 13-item example in 26 fixed item orders.

The circuit lemma is checked independently by enumerating every basis family that extends a fixed circuit of each size from two through five by one nonloop item, testing the basis-exchange axiom, and computing complete rank tables. The checker then forms the missing-pair graph from rank increments and tests its multipartite structure and every own-color case for the first later item. A separate rank-three check enumerates every side-line membership pattern allowed by rank submodularity. The few-label argument is checked on all coarsenings of the example's labels into three classes or the unlabelled core. The general bounds and minimality results are proved in Sections 4–7; these finite checks are supporting evidence, not an exhaustive enumeration of all matroids at those ranks.

The separate singleton-profile checker compares the localized-pair algorithm with every original canonical basis. It covers the zero-extra case, the sharp represented examples, small arbitrary matroids, and coarsenings and seeded orders of the nonrepresentable example. It checks the quotient equivalence even when contracted basis items fail, promotes original-basis items independently, and verifies every decreasing-potential step. A separate enumeration checks all graphs reachable by protected or unprotected binary cuts on complete graphs of sizes two through six.

Its block decomposition is computed from graph edges, without assuming the clique invariant. Both checkers refuse optimized Python mode that would disable assertions.

All algorithmic requirements are explicit: the original independence oracle, the finite label lists and the fixed item order. The probability calculations are proof devices, and the code uses only exact integer arithmetic and Python's standard library.

Part II

The rank-four obstruction and its amplification

II.1 Statement and scope

The sharp rank-only bound $r\binom{h+d}{d}$ does not extend to fixed-cost summaries of arbitrary oracle matroids. It already fails for rank four, one label per item, and two failed labels.

Theorem 1. There is a rank-four matroid on 13 original items, each carrying one of four labels and a fixed positive integer cost, such that every original-ground restriction preserving the minimum cost of a surviving basis after every at-most-two-label failure must retain all 13 items. All six two-label queries have surviving rank four. In contrast, $r\binom{h+d}{d} = 4\binom{3}{1} = 12$.

Consequently 12 items cannot suffice even for preserving only the exact minimum-cost value; preserving every cost prefix or the canonical optimum is at least as demanding.

Theorem 2. For every integer $h \geq 2$, there is a rank-four, one-label-per-item example requiring

$$4(h+1)+1 \tag{WO1}$$

original items. Taking direct sums gives, for every rank r divisible by four, examples requiring $r(h+1)+r/4$. Thus the one-failure weighted oracle theorem is a substantive endpoint. The represented-matroid theorem is unaffected: the resource matroid used here is not representable over any field, as verified directly in Section 6.

These are exact finite constructions with proofs. They do not assert an optimal general weighted-oracle bound above the displayed obstruction. The two-failure upper-bound companion proves that every rank-at-most-three matroid, and every instance with at most three occurring labels, still admits a $3r$ prefix summary at $h = 2, d = 1$. Thus Theorem 1 has the smallest possible rank, label count and original ground size for a violation in that fixed model. The same companion proves that the universal rank-four maximum is exactly 13 and gives an exact one-failure-profile certificate for this instance. It also proves that a violation requires at least two distinct items outside the original basis in the one-failure union; this example has exactly two.

II.2 The underlying eight-class matroid

Let the underlying class ground be $\{0, 1, \ldots, 7\}$. Declare every set of at most three classes independent, and declare a four-class set a basis except for the following five forbidden sets:

$$
\begin{aligned}
&H_1 = \{0,1,2,3\}, \quad H_2 = \{0,1,4,5\},\\
&H_3 = \{0,1,6,7\}, \quad H_4 = \{2,3,4,5\},\\
&H_5 = \{2,3,6,7\}.
\end{aligned}
\tag{WO2}
$$

Sets larger than four are dependent. This is a relabeling of the classical Vámos matroid. Its needed properties follow from the explicit definition; no representation theorem is assumed.

Lemma 2.1. The declared bases define a rank-four matroid.

Proof. Different forbidden four-sets intersect in at most two classes. Take two allowed bases B, C and $e \in B \setminus C$. If $|C \setminus B| = 1$, its only replacement gives C itself. Otherwise the candidate sets $B - e + f$, for $f \in C \setminus B$, are distinct four-sets sharing the three-set $B - e$. At most one can be forbidden, so an allowed exchange exists. This is the basis-exchange axiom.

Every three-set has five possible fourth classes, and at most one of its four-set extensions is forbidden. Thus every set of at most three classes is contained in an allowed basis, as declared. There are $\binom{8}{4} - 5 = 65$ bases, so the rank is four. □

Replace some underlying classes by parallel copies. A set of copies is independent exactly when it contains at most one copy of each class and its class set is independent in WO2. This defines a matroid: heredity is immediate, and augmentation follows by augmenting the smaller underlying independent class set with a class from the larger one and then taking its available copy. This explicit parallel-extension oracle is the entire resource interface below.

It is an explicit, efficiently evaluable oracle: check duplicate classes, the cardinality, and the five constant-size forbidden sets. The lower bound concerns the retained original ground, not computational difficulty hidden inside the oracle.

II.3 The 13 ordered, colored items

Give item i cost $c_i = 2^{i-1}$. All subset costs are distinct, so each surviving optimum is unique. The cost order is the item order in this table. The last column gives one failure query under which that item is forced into the unique optimum.

Item i	Underlying class	Its only label	A necessity witness
1	5	B	fail C,D
2	2	A	fail C,D
3	3	A	fail C,D
4	1	C	fail B,D
5	7	D	fail B,C
6	4	B	fail A,D
7	0	C	fail A,B
8	6	D	fail A,C
9	2	D	fail A,B
10	1	B	fail C,D
11	5	C	fail B,D
12	4	D	fail B,C
13	2	C	fail A,D

Every underlying class has a copy, so the extension has rank four. Failure of a label removes every item carrying that label. The six two-label optima are as follows.

Failed labels	Surviving labels	Unique minimum-cost basis	Cost
A,B	C,D	$\{4, 5, 7, 9\}$	344
A,C	B,D	$\{1, 5, 6, 8\}$	177
A,D	B,C	$\{1, 4, 6, 13\}$	4137
B,C	A,D	$\{2, 3, 5, 12\}$	2070
B,D	A,C	$\{2, 3, 4, 11\}$	1038
C,D	A,B	$\{1, 2, 3, 10\}$	519

Lemma 3.1. This table lists the canonical greedy basis and the unique minimum-cost basis in every displayed restriction.

Proof. Greedy scans the surviving items in the given increasing cost order. It rejects repeated underlying classes and the five forbidden four-sets in WO2. The complete nontrivial rejection traces are short:

- With labels C,D surviving, greedy takes $4, 5, 7$, rejects item 8 because its classes would be H_3, and takes 9.
- With labels B,D surviving, it takes $1, 5, 6, 8$. The four classes are $\{4, 5, 6, 7\}$, the allowed union of the last two underlying pairs.
- With labels B,C surviving, it takes $1, 4, 6$, rejects item 7 by H_2, rejects 10 and 11 as parallel repetitions, and takes 13.
- With labels A,D surviving, it takes $2, 3, 5$, rejects 8 by H_5 and 9 as a repetition, and takes 12.
- With labels A,C surviving, it takes $2, 3, 4$, rejects 7 by H_1, and takes 11.
- With labels A,B surviving, it takes $1, 2, 3$, rejects 6 by H_4, and takes 10.

All listed accepted four-sets are allowed bases. Matroid greedy minimizes the fixed additive cost of a basis, and distinct powers of two make this optimum unique. The numerical costs are the corresponding sums. □

Proof of Theorem 1. The union of the six displayed bases is all 13 items. If a restriction omits item i, use its necessity query from the first table. The unique original optimum contains i, so the restricted instance either has no rank-four basis or has strictly larger optimum cost. Either outcome violates the prescribed guarantee. Thus every valid restriction retains all 13 items. This also proves the prefix obstruction: at the prefix ending in an accepted item, omitting that item loses the greedy rank increase. □

II.4 Amplification to every failure budget at least two

Fix $h \geq 2$. Before the 13-item order, prepend $h - 2$ blocks $Q_1, \ldots, Q_{h-2}$. Each block consists of four new parallel copies whose underlying classes are the allowed basis $\{1, 2, 3, 5\}$. Give all four copies in Q_j a new single label Z_j; these labels are distinct and do not belong to the original four-label alphabet. Assign costs as successive powers of two in the complete new order.

Every item of Q_j is forced by the query failing $Z_1, \ldots, Z_{j-1}$. The remaining earlier blocks are absent, and Q_j is the first surviving block and a full basis. Greedy takes all of it. This query uses at most $h - 3$ failures whenever such a block exists.

Every one of the original 13 items is still forced: fail all $h - 2$ new block labels, then use its two-label witness from Section 3. This uses exactly h failures and leaves the same ordered original instance. There are $4(h - 2) + 13 = 4(h + 1) + 1$ forced items and the resource rank remains four. This proves WO1.

In fact every allowed failure still leaves rank four. If a new block survives, it is a full basis. If no new block survives, at least $h-2$ failures were spent on their labels, leaving at most two failures among the four original labels. Section 3 then supplies a surviving basis.

For t disjoint copies of the construction, take the direct sum of their resource matroids and reuse the same label alphabet across copies. A surviving basis is a union of bases of the separate components. Give all items distinct power-of-two costs, keeping the prescribed local order inside each component. The optimum is unique and componentwise optimal, so the same witness still forces each copy of each item. The rank is $r = 4t$, and the necessary support is $t(4(h+1)+1) = r(h+1) + r/4$. □

II.5 Why this does not contradict the rank-only theorem

The nine-item restriction

$$H_{\text{rank}} = \{1, 2, 3, 5, 7, 10, 11, 12, 13\} \tag{WO3}$$

already preserves full rank after every allowed failure in the base 13-item example. For failures A,B; A,C; A,D; B,C; B,D; and C,D, respectively, it contains the following surviving bases:

$$\begin{array}{ccc} \{5,7,11,12\}, & \{1,5,10,12\}, & \{1,7,10,13\}, \\ \{2,3,5,12\}, & \{2,3,7,11\}, & \{1,2,3,10\}. \end{array}$$

Their underlying class sets are all allowed by WO2. Every zero- or one-label failure leaves a superset of one of these two-label surviving instances, so it also retains rank four. Some of these bases are more expensive than the true optimum, which is permitted by rank preservation.

Thus the sharp rank-only summary, the one-failure weighted oracle summary, and the represented fixed-cost theorem have different valid scopes. The obstruction rules out silently transferring the last theorem to all independence-oracle matroids for $h \geq 2$.

II.6 Nonrepresentability and source attribution

Here is a short independent check of the representation boundary. Group the underlying classes into pairs $P = \{0,1\}$, $Q = \{2,3\}$, $R = \{4,5\}$, and $S = \{6,7\}$. Suppose vectors over a field represented WO2, and let the same letters denote their two-dimensional spans. The five forbidden pair unions have dimension three, whereas $R+S$ has dimension four. Also $P+Q+R$ and $P+Q+S$ have dimension four.

Since $\dim(P+Q) = 3$, let $L = P \cap Q$, a line. Both $R \cap P$ and $R \cap Q$ are lines. If they were different, they would span R inside $P+Q$, contradicting $\dim(P+Q+R) = 4$. They must therefore be the same line, namely L, so $L \subseteq R$. The identical argument gives $L \subseteq S$. This forces $\dim(R+S) \leq 3$, contradicting its required dimension four. No field representation exists. A parallel extension containing every class cannot remove this obstruction.

The underlying matroid is classical; the 13-item coloring, order, failure table, and amplification above are the concrete obstruction used here. For primary background on the Vámos matroid, see Karn, Nasr, Proudfoot and Vecchi, *Equivariant Kazhdan–Lusztig theory of paving matroids*, Section 5. None of its representation or polynomial-invariant results is needed for the proof above. No publication-priority claim is made for this failure-summary application.

II.7 Reproducible exact certificate

Run

```
python3 -B \
  problems/matroid-failure-summaries/verify/check_weighted_obstruction.py
```

The checker independently enumerates the 65 underlying bases, the 386 extended bases, and all 8,192 subsets of the 13-item ground. It verifies basis exchange, constructs the complete rank table by exhaustive subset recurrence, and compares greedy with literal minimum cost over all surviving bases. It checks all zero-, one-, and two-label queries, every item's optimum and prefix necessity witness, and the rank-only restriction WO3. Small amplified instances are also checked directly; the proof in Section 4 establishes the unbounded family.

The computational search used to discover the example is not part of this certificate or a premise of the theorem. The committed checker is deterministic and verifies only the fully specified construction.

Attribution and review scope. Yilin Liu and GPT-6 Astra Pro. *Algorithm Frontier*, 26 October 2026. These are exact independence-oracle statements. The finite checkers test independent rank tables, original-prefix necessity, quotient identities and the specified constructions; they do not replace the universal arguments. External peer review and proof-assistant formalization are not claimed.